Sri Chaitanya ACADEMY | Infinity Learn



JEE CHAMPION TEST SERIES

JEE Main Full Syllabus Test - 5

Test Date: 12th Mar 2025

INSTRUCTIONS

- The test is of 3 hours duration.
- The Test Booklet consists of 75 questions. The maximum marks are 300.
- There are three parts in the question paper (1, 2, 3) consisting of Physics, Chemistry and Mathematics, having 25 questions in each subject and each subject having two sections.
  (I) Section-A contains 20 multiple choice questions with only one correct option.
  (II) Section-B contains 5 numerical value type questions.
- Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.
- The answer should be within 0 to 9999. If the answer is in decimal, then round off to the nearest integer value (e.g., If the answer is above 10 and less than 10.5, round off to 10; if the answer is more than or equal to 10.5 but less than 11, round off to 11).
- Use blue/black ballpoint pen only for writing particulars or marking responses on the answer sheet. Use of pencil is strictly prohibited.

PHYSICS

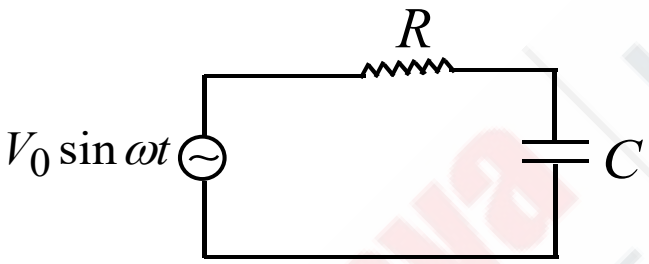
Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. An ac voltage source $V = V_{0}\sin\omega t$ is connected across resistance R and capacitance C as shown in figure. It is given that $R = \frac{1}{\omega C}$. The peak current is I_0. If the angular frequency of the voltage source is changed to $\frac{\omega}{\sqrt{3}}$ keeping R and C fixed, then the new peak current in the circuit is



1) $\frac{I_0}{2}$ **2)** $\frac{I_0}{\sqrt{2}}$ **3)** $\frac{I_0}{\sqrt{3}}$ **4)** $\frac{I_0}{3}$

2. **Assertion:** Thin films such as soap bubble or a thin layer of oil on water show beautiful colours when illuminated by white light.

Reason: It happens due to the interference of light reflected from only the upper surface of the thin film.

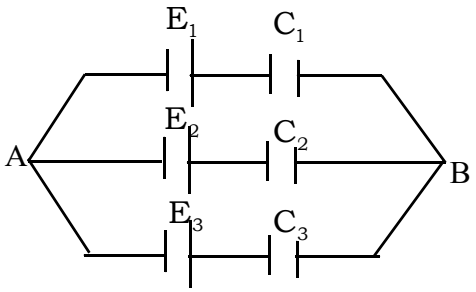
1) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion

2) If both Assertion and Reason are true but Reason is not correct explanation of the Assertion

3) If Assertion is true but Reason is false

4) If Assertion is false but the Reason is true

3. In the given circuit find the $\left|V_A - V_B\right|$ where V_A, V_B are electric potentials at the points A and B



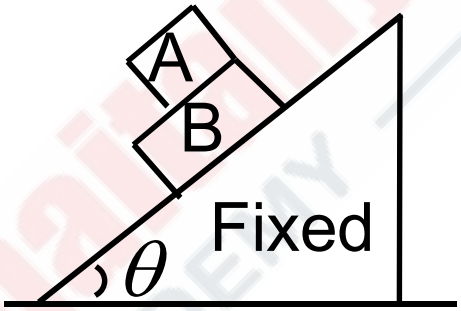
1) $\frac{E_1 + E_2 + E_3}{3}$

2) $\frac{3}{\frac{1}{E_1} + \frac{1}{E_2} + \frac{1}{E_3}}$

3) $\frac{E_1C_1 + E_2C_2 + E_3C_3}{C_1 + C_2 + C_3}$

4) $\frac{\frac{E_1}{C_1} + \frac{E_2}{C_2} + \frac{E_3}{C_3}}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$

4. A block A slides over another block B which is placed over a smooth inclined plane as shown in figure. The coefficient of friction between the two blocks A and B is μ. Mass of block B is two times the mass of block A. The acceleration of the centre of mass of two blocks is



1) $g\sin\theta$

2) $\frac{g\sin\theta - \mu g\cos\theta}{3}$

3) $\frac{g\sin\theta}{3}$

4) $\frac{2g\sin\theta - \mu g\cos\theta}{3}$

5. **Statement-I:** Due to rapid pumping of types, air inside the tyre is hotter than atmospheric air

Statement-II: Adiabatic process occurs at very high rate

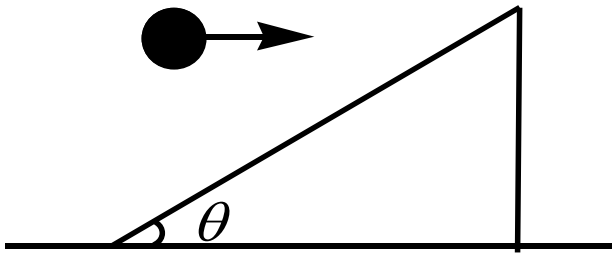
1) Statement-I is true; Statement-II is false

2) Statement-I is false; Statement-II is true;

3) Statement-I & Statement-II are true

4) Statement-I & Statement-II are false

6. An inclined plane is placed on a horizontal smooth surface. The plane is struck by an elastic ball whose velocity is horizontal just before the impact. The ball bounces off the inclined plane and then lands on it again at the point of first impact. Find the ratio of the masses of the ball and the inclined plane. (Angle $\theta = 30^\circ$)



1) 2 **2)** 3 **3)** 4 **4)** 5

7. **Assertion** The resistivity of a semiconductor increases in temperature.

Reason In a conducting solid, the rate of collisions between free electrons and ions increases with increase of temperature.

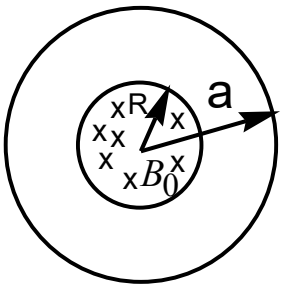
1) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion

2) If both Assertion and Reason are true but Reason is not correct explanation of the Assertion

3) If Assertion is true but Reason is false

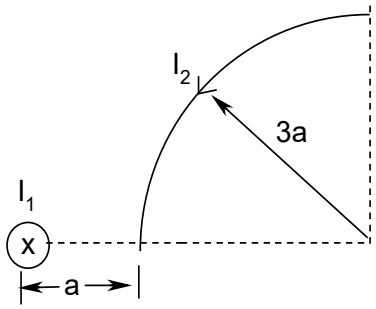
4) If Assertion is false but the Reason is true

8. A uniform magnetic induction field B_0 exists in a cylindrical region of radius R. A conducting ring of radius a (>R) is placed co-axially with the field region. The ring has mass 'm' and linear charge density λ . The angular velocity gained by the ring if the field is switched off is



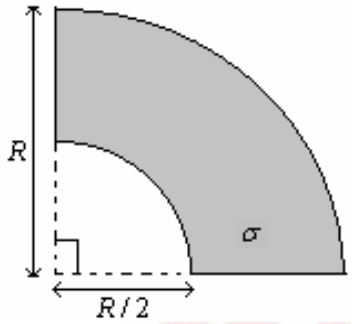
1) $\frac{\lambda\pi B_0 a^2}{mR}$ **2)** $\frac{\lambda\pi B_0 a^2}{2mR}$ **3)** $\frac{\lambda\pi B_0 R^2}{ma}$ **4)** $\frac{\lambda\pi B_0 R^2}{2ma}$

9. A long straight wire carries current I_1. There is quarter ring of radius 3a present near it as shown. Find force on the quarter ring. Quarter ring is in the plane of paper and long wire is perpendicular to the plane of paper.



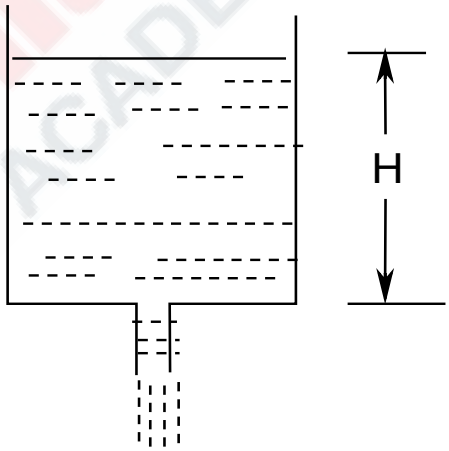
1) $\frac{\mu_0 I_1 I_2}{\pi}\ln$ **2)** $\frac{\mu_0 I_1 I_2}{2\pi}\ln$ **3)** $\frac{\mu_0 I_1 I_2}{\pi}\ln$ **4)** $\frac{\mu_0 I_1 I_2}{2\pi}\ln 2$

10. The diagram shows A part of disc of radius R carrying uniformly distributed charge of density σ. Electric potential at the centre O of parent disc is



1) $\frac{\sigma R}{2\varepsilon_0}$ **2)** $\frac{\sigma R}{16\varepsilon_0}$ **3)** $\frac{\sigma R}{24\varepsilon_0}$ **4)** $\frac{\sigma R}{32\varepsilon_0}$

11. A tank of cross sectional area A_0 is filled with a liquid. A small orifice of area $A\left(A << A_0\right)$ is present at the bottom of tank. At a moment the height of liquid in the tank is 'H', what is acceleration of top layer of liquid at this moment?



1) g **2)** $\frac{A}{A_0}g$ **3)** $\left(\frac{A}{A_0}\right)^2 g$ **4)** depends on H

12. Statement 1: A metallic surface is irradiated by a monochromatic light of frequency $v > v_0$ (the threshold frequency). The maximum kinetic energy and the stopping potential are K_{max} and V_0 respectively. If the frequency incident on the surface is doubled, both

the K_{max} and V_0 are also doubled.

Statement 2: The maximum kinetic energy and the stopping potential of photoelectrons emitted from a surface are linearly dependent on the frequency of incident light

1) Statement 1 is True, Statement 2 is False.

2)Statement 1 is True, Statement 2 is True; Statement 2 is a correct explanation for Statement 1.

3)Statement 1 is False, Statement 2 is True.

4)Statement 1 is True, Statement 2 is True; Statement 2 is not the correct explanation for Statement 1.

13. A certain quaternary star system consists of three stars, each of mass m, moving in same circular orbit about a stationary central star of mass M. The three identical stars orbit in same sense and are symmetrically located with respect to each other (The centre of all stars lie in one plane). Considering gravitational force of all remaining bodies on every star, the time period of each of three stars is

1) $2\pi\sqrt{\dfrac{r^3}{G\left(M+\dfrac{m}{3}\right)}}$ **2)** $2\pi\sqrt{\dfrac{r^3}{G\left(M+\dfrac{m}{\sqrt{3}}\right)}}$

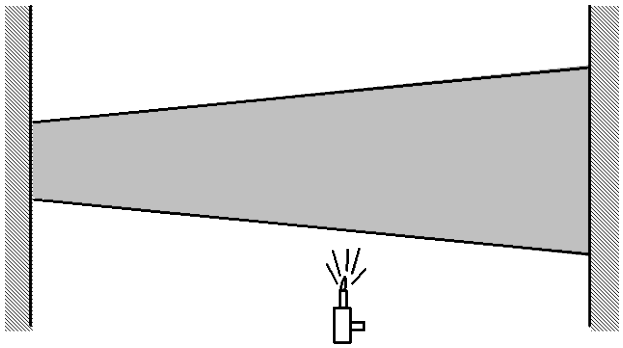
3) $2\pi\sqrt{\dfrac{r^3}{G(M+3m)}}$ **4)** $2\pi\sqrt{\dfrac{r^3}{G\left(M+\sqrt{3}m\right)}}$

14. A rod of length ℓ and cross section area A has a variable thermal conductivity given by $k=\alpha T$, where α is a positive constant and T is temperature in kelvin. Two ends of the rod are maintained at temperatures T_1 and $T_2\left(T_1>T_2\right)$. Heat current flowing through the rod will be

1) $\dfrac{A\alpha\left(T_1^2-T_2^2\right)}{\ell}$ **2)** $\dfrac{A\alpha\left(T_1^2-T_2^2\right)}{4\ell}$

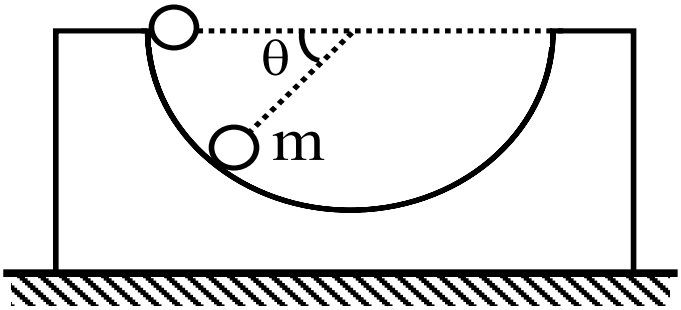
3) $\dfrac{A\alpha\left(T_1^2-T_2^2\right)}{3\ell}$ **4)** $\dfrac{A\alpha\left(T_1^2-T_2^2\right)}{2\ell}$

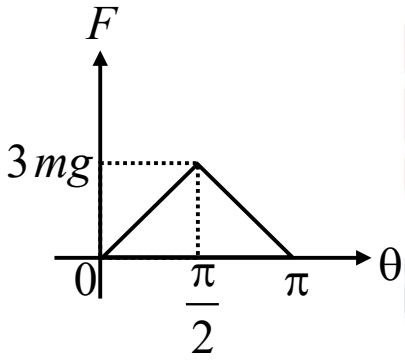
15. A rod is made of uniform material and has non-uniform cross section. It is fixed at both the ends as shown and heated at mid-section. Which of the following statement is correct?

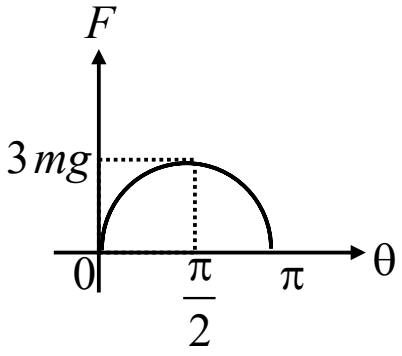


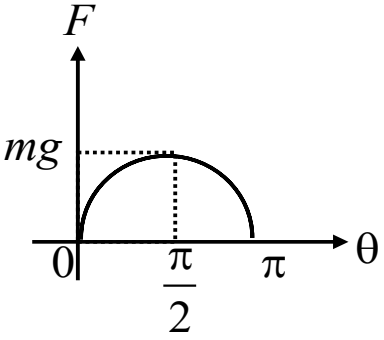
1) Force of compression in the rod will be maximum at mid-section
2) Compressive stress in the rod will be maximum at left end
3) Since rod is fixed at both the ends, its length will remain unchanged. Hence, no strain will be induced in it
4) None of these

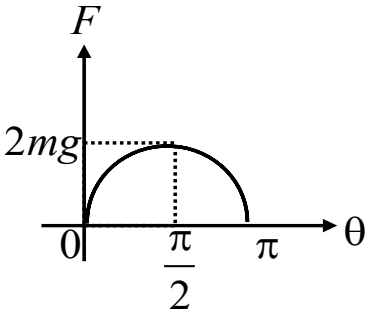
16. In the shown fig., small mass m starts sliding down a smooth and stationary circular tack. Which of the following graph best represents the variation of magnitude of the force applied by the track on the mass and the angle θ:



1)


2)


3)


4)


17. A linearly polarized electromagnetic wave given as $E = E_{\circ}\hat{i}\cos(kz - \omega t)$ is incident normally on a perfectly reflecting infinite wall at $z = a$. Assuming that the material of the wall is optically inactive, the reflected wave will be given as

1) $E_r = -E_\circ \hat{i} \cos(kz - \omega t)$ **2)** $E_r = E_\circ \hat{i} \cos(kz + \omega t)$

3) $E_r = -E_\circ \hat{i} \cos(kz + \omega t)$ **4)** $E_r = E_\circ \hat{i} \sin(kz + \omega t)$

18. The length of a metal wire is l_1 when the tension in it is T_1 & l_2 when the tension is T_2. The natural length of the wire is

1) $\frac{l_1T_2 - l_2T_1}{T_2 - T_1}$ **2)** $\frac{l_1T_2 - l_2T_1}{T_1 + T_2}$ **3)** $\frac{l_1T_2 - l_2T_1}{T_1 - T_2}$ **4)** $\frac{l_1T_1 - l_2T_2}{T_1 + T_2}$

19. The following equations represent transverse waves

$z_1 = A\cos(3Kx - \omega t)$ …(i)

$z_2 = A\cos(4Ky - \omega t)$ …(ii)

A wave is formed by the superposition of waves (i) and (ii). This wave is

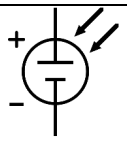
1) Traveling in the direction making an angle $\tan^{-1}\frac{3}{4}$ with +ve x–axis

2) Traveling in the direction making an angle $\tan^{-1}\frac{4}{3}$ with +ve x–axis

3) Traveling in the direction making an angle $\pi - \tan^{-1}\frac{3}{4}$ with +ve y–axis

4) Traveling in the direction making an angle $\pi + \tan^{-1}\frac{4}{3}$ with +ve y–axis

20. Match the following Column I with Column II

	Column I		**Column II**
I	Photodiode	A	
II	Solar cell	B	
III	Light emitting diode	C	
IV	Zener diode	D	

1) I-A, II-B, III-C, IV-D **2)** I-D, II-A, III-B, IV-C

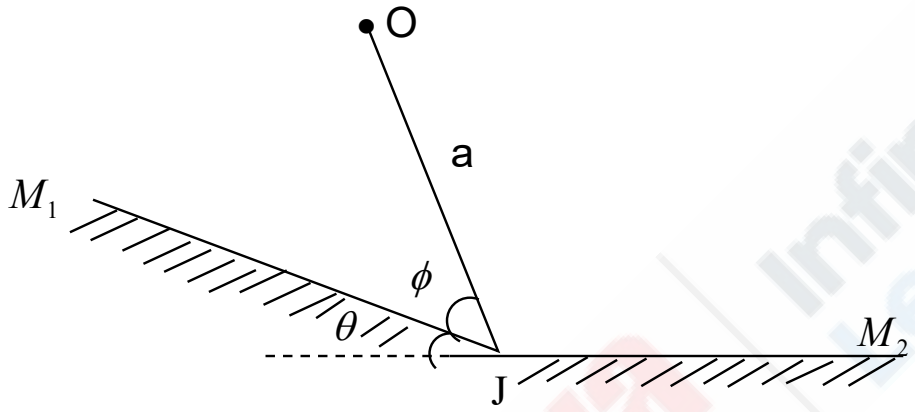
3) I-C, II-D, III-B, IV-A **4)** I-B, II-A, III-C, IV-D

(NUMERICAL VALUE TYPE)

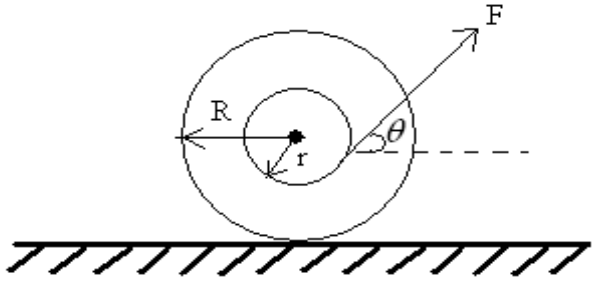
Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only. First 5 attempted questions will be considered if more than 5 questions attempted. The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if **not attempt** and **-1** in all other cases.

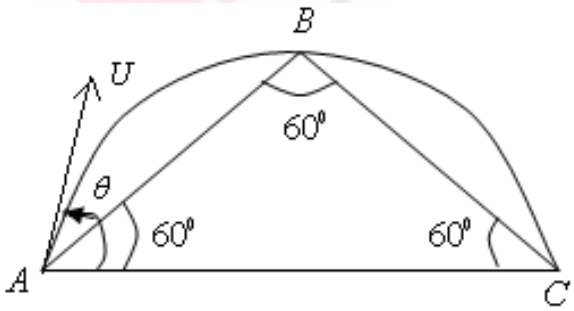
21. Two mirrors M_1 and M_2 are inclined at angle θ as shown. An object O is placed at a distance 'a' from the joint J of mirrors. Distance between first image formed by each of the mirrors is na, then n is _____ $(\phi = 60^0 \& \ \theta = 30^0)$



22. The spool shown in the figure is placed on a rough horizontal surface has inner radius r and outer radius R. The angle θ between the applied force F (tangential to the circle of radius r) and the horizontal can be varied. The critical angle (θ) for which the spool does not roll and remains stationary is $\frac{\pi}{n}$, then n is___(if $R = 2r$)



23.

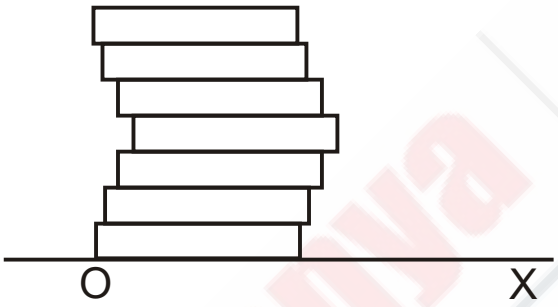


ABC is an equilateral triangle. From point A projectile was thrown such that it just grazes over B and reaches C. Given length of $AB = l$. The angle of projection 'θ' is $\tan^{-1}\left(n\sqrt{3}\right)$, then n is ____

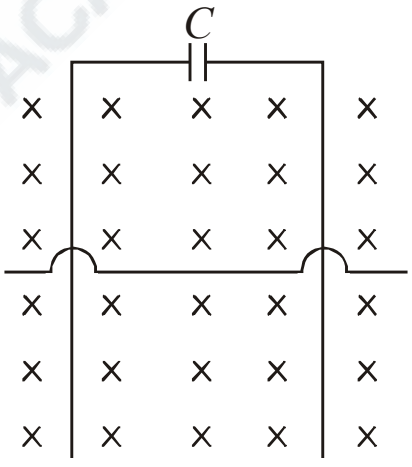
24. An $\alpha - particle$ with K.E. T= 5.3 Mev initiates a nuclear reaction $Be^9 +_2 \alpha^4 \rightarrow C^{12} +_0 n^1$ with energy yield Q=+5.7 Mev. Then the K.E. of neutron outgoing at the right angle to the motion direction of the $\alpha - particle$ will be [initially Be^9 is at rest] $\frac{68}{n}$ (in MeV), then n is

____

25. Seven homogeneous bricks, each of length *L*, are arranged as shown in figure. Each brick is displaced with respect to the one in contact by *L*/10. The x-coordinate of the centre of mass relative to the origin O shown is $\frac{22}{5n}L$. Find the value of n.



26. Two metal bars are fixed vertically and are connected on the top by a capacitor C. A sliding conductor of length ℓ and mass m slides with its ends in contact with the bars. The arrangement is placed in a uniform horizontal magnetic field directed normal to the plane of the figure. The conductor is released from rest. Find the displacement of the conductor (in metre) after 2s. Assume that $CB^2\ell^2 = 4m$ (the mass of the conductor) and $g = 10m/s^2$.

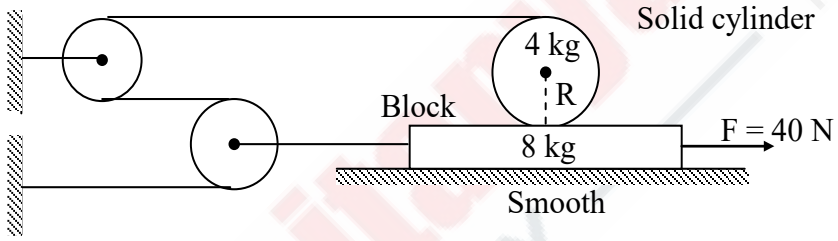


27. A highly accurate determination of *g* has a quoted error of 6 parts in 10^9. If the dependence of *g* on geographical latitude at sea level is given by $g = g_0\left(1+\beta\sin^2\phi\right)$ where ϕ is the

latitude and β is a dimensionless constant with a value of 0.0055.The northward displacement near latitude 45^0 which gives a change in *g* equal to the quoted error (radius of earth $= R = 6.4 \times 10^6 m$) is (0.1)n m, where n is_____(approximately)

28. A stationary He^+ ion emitted a photon corresponding to the first line of the Lyman series. The photon liberates electron from a stationary hydrogen atom in the ground state. The velocity of the liberated electron is 3.1×10^n m/s. Find n (You can make necessary approximations)

29. An ideal string is wrapped several times on a solid cylinder of mass 4 kg and radius 1 m. The pulleys are ideal and the surface between block and ground is smooth. If the torque acting on the cylinder is $\frac{80}{n}$ N-m, then find the value of n.



30. In the standard young double sit experimental set up the fringe width is found to be 'β'. The minimum distance of a point on the screen from the zeroth maxima where the intensity is 3/4th of that at a maxima is $\frac{\beta}{n}$. Find the value of n

CHEMISTRY

Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

31. Assertion: Maltose & Lactose both are reducing sugars

Reason: Carbohydrates having free – OH group at anomeric carbon can reduce Tollen's reagent

1) Both Assertion (A) and Reason (R) are the true and Reason (R) is a correct explanation of Assertion (A)

2) Both Assertion (A) and Reason (R) are the true but Reason (R) is not a correct explanation of Assertion (A)

3) Assertion (A) is true and Reason (R) is false

4) Assertion (A) is false and Reason (R) is true.

32. Assertion: For Hydrogen atom degeneracy of 3rd level is 3

Reason: Aufbau principle is not applicable for s, p, d, f subshells of same shell for single electronic species

1) Both Assertion (A) and Reason (R) are the true and Reason (R) is a correct explanation of Assertion (A)

2) Both Assertion (A) and Reason (R) are the true but Reason (R) is not a correct explanation of Assertion (A)

3) Assertion (A) is true and Reason (R) is false

4) Assertion (A) is false and Reason (R) is true.

33. Statement -1: common name of phenol is Carbolic acid

Statement-2: phenol gives effervescence with $aqNaHCO_3$

Statement-3: phenol is more acidic than ethanol

Statement-4: phenol is more acidic than picric acid

1) 1 and 3 are correct **2)** 1, 2 and 3 are correct

3) 2, 3 and 4 are correct **4)** 2 and 4 are correct

34. **Statement-1:** Emf of a cell is an extensive property

Statement-2: Larger cells (AA batteries) have more emf than smaller cells (AAA batteries)

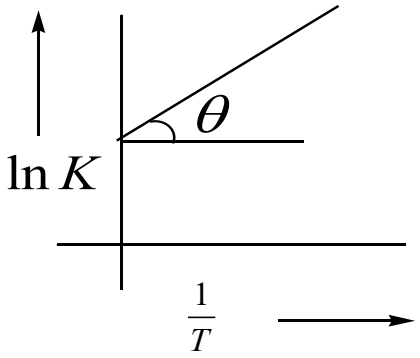
Statement-3: Fe cannot be used to protect Zn in Galvanization

Statement-4: For concentration cell $E°_{Cell} = 0$

1) 1 and 2 are correct **2)** 1 and 3 are correct

3) 1 and 4 are correct **4)** 3 and 4 are correct

35. The correct statement is

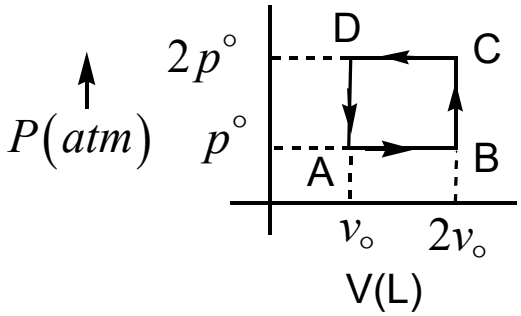


(K=Equilibrium Constant, T=Temperature)

1) Given reaction is endothermic **2)** Given reaction is exothermic

3) Given reaction has $\Delta S^\circ < 0$ **4)** Given reaction has $K_2 > K_1$ for $T_2 > T_1$

36. The work done in the process ABCD is

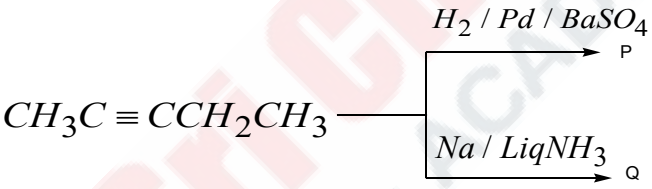


1) $p^\circ v_\circ$ **2)** $-p^\circ v_\circ$ **3)** $2p^\circ v_\circ$ **4)** $-2p^\circ v_\circ$

37. Which of the following complex ion has SP^3 hybridized metal centre?

1) $[Ni(CN)_4]^{2-}$ **2)** $[Ni(CO)_4]$ **3)** $[PdCl_4]^{2-}$ **4)** $[Pt(CN)_4]^{2-}$

38.



The Correct statement is

1) P is a trans alkene

2) Q is a cis alkene

3) P is a mixture of cis and trans Alkenes

4) P and Q both have non zero dipole moment

39. Hex-4-en-2-ol on treatment with PCC gives 'A'. A on reaction with sodium hypoiodite gives 'B', which on further heating with soda lime gives 'C'. The compound 'C' is:

1) 3-hexene **2)** Butanaldehyde

3) 2-butene **4)** 4-methyl pent-2-ene

40. Reaction of Butanamide with $Br_2 / KOH(aq)$ produces

1) Ethyl nitrile **2)** Propyl amine **3)** Propane nitrile **4)** Ethylamine

41. Primary, Secondary and tertiary amines can be separated using

1) Para-Toluene Sulphonyl Chloride **2)** Acetic Acid

3) Chloroform and KOH **4)** dil HCl

42. Complete reduction of benzene diazonium Chloride with Zn/HCl gives:

1) Aniline **2)** Phenyl Hydrazine

3) Azobenzene **4)** Hydrazobenzene

43. If one strand of a DNA has the sequence ATGCTTCA, sequence of bases in complementary strands is _____

1) CATTAGCT **2)** TACGAAGT

3) GTACTTAC **4)** ATGCGACT

44. Consider the oxides of group 14 elements $SiO_2, GeO_2, SnO_2, PbO_2, CO$ & GeO. The amphoteric oxides are

1) GeO, GeO_2 **2)** SiO_2, GeO_2 **3)** SnO_2, CO **4)** SnO_2, PbO_2

45. Which of the following element will remain as liquid inside pure boiling water

1) Ca **2)** Ga **3)** Li **4)** Br

46. The sum of Covalency and Oxidation state of boron in $(BF_4)^-$ are

1) 7 **2)** 8 **3)** 9 **4)** 10

47. Identify the incorrect pair

1) Borax - $Na_2B_4O_7.10H_2O$ **2)** Cryolite - Na_3AlF_6

3) Fluorapatite - $3Ca_3(PO_4)_2.CaF_2$ **4)** Fluorspar - BF_3

48. The correct order of Bond dissociation enthalpy is:

1) $Cl_2 > Br_2 > F_2 > I_2$ **2)** $F_2 > Cl_2 > Br_2 > I_2$

3) $Cl_2 > F_2 > Br_2 > I_2$ **4)** $I_2 > Br_2 > Cl_2 > F_2$

49. On adding small quantity of naphthalene, the boiling point of benzene will

1) Increase **2)** remains same

3) first decrease and then increases **4)** decrease

50.

COLUMAN –I Species		**COLUMNII-II Dipole moment (D)**	
1	HF	p	0.23
2	H_2O	q	1.47

3	NH_3	r	1.78
4	NF_3	s	1.85

Correct match is

1) $1\to s,2\to q,3\to s,4\to p$ **2)** $1\to r,2\to s,3\to q,4\to p$

3) $1\to p,2\to q,3\to r,4\to s$ **4)** $1\to s,2\to q,3\to r,4\to p$

(NUMERICAL VALUE TYPE)

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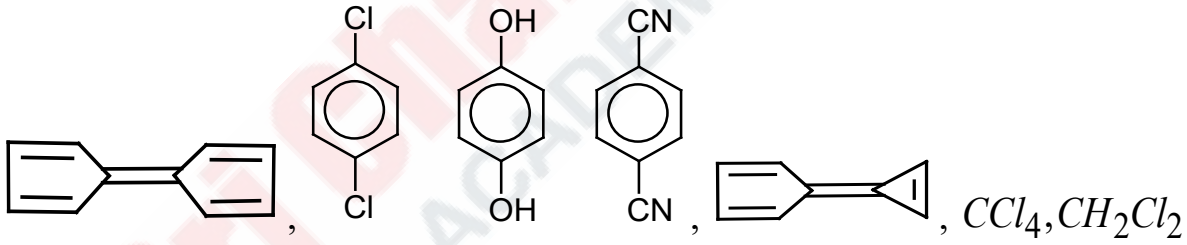
Marking scheme: +4 for correct answer, 0 if **not attempt** and **-1** in all other cases.

51. The value of $\Delta G^{\circ}\left(in\,KJ\right)$ for the conversion of 3 moles of oxygen to ozone at 298K is_____ (Kp for the reaction $\left(\frac{3}{2}O_2(g)\to O_3(g)\right)$ is 2.5×10^{-29} & $\log 2=0.3$

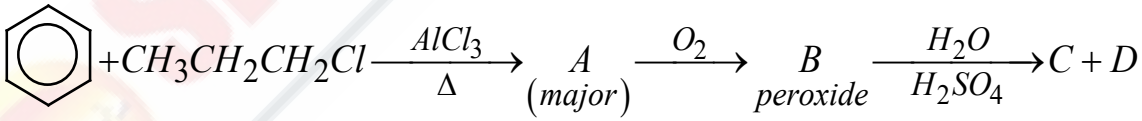
$\ln x=2.3\log x$ & $R=8.3Jmol^{-1}K^{-1}$

52. The total number of electrons present in HOMO (Highest Occupied Molecular Orbital) of O_2 is _____

53. Number of polar molecules $\left(\mu\neq0\right)$ among the following is _____



, , CCl_4,CH_2Cl_2

54. 

C gives Iodoform test. Then molecular weight of D in (g) is_____

55. A first order reaction is completed by 99.9% in 90 days. Half-life of the reaction (in days) is_____ $(\log 2=0.3)$

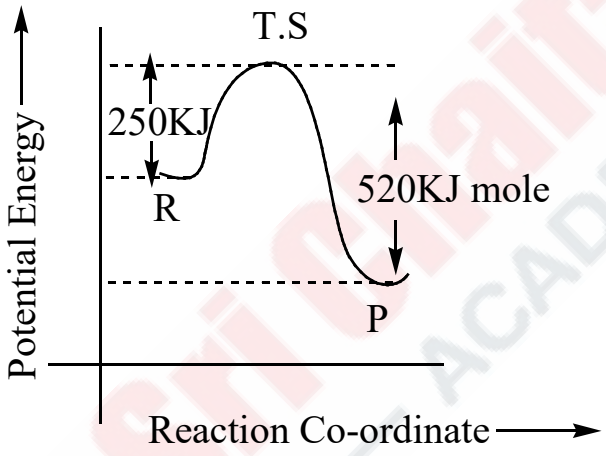
56. P^H of resultant solution formed when 100 ml of 0.1M H_3PO_4 reacts with 100 ml of 0.1 M $NaOH$ is _____ $\left(Ka_1H_3PO_4 = 2.5\times10^{-3}, Ka_2 = 10^{-8}, Ka_3 = 10^{-13}\right)$

57. The difference in number of electrons present in largest and smallest species among Mg, Mg^{2+}, Al, Al^{3+} is_____

58. Density of 3M solution of $NaCl$ is $1.25g/mL^{-1}$. The molality of solution is_____ moles kg^{-1} (at wt of Na = 23, Cl = 35.5)

59. The number of ions which will be precipitated by H_2S in acidic medium among $\left(Cu^{2+}, Ni^{2+}, Mn^{2+}, Pb^{2+}, Co^{2+}Al^{3+}, Hg^{2+}\right)$ is_____

60. If the reaction comes to equilibrium at 300 K then the magnitude of $\Delta S(inJK-1)$ is_____



MATHEMATICS

Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

61. The value of $\tan\left(\sum_{r=1}^{\infty}\cot^{-1}\left(\frac{r^2}{2}+\frac{15}{8}\right)\right)$ is equal to

1) 1 **2)** 2 **3)** 3 **4)** 4

62. Consider the equation $\frac{xy}{x+y}=2^2 3^4 5^6$ then the number of ordered pairs (x,y) where $x,y\in N$ satisfying the above equation are

1) 585 **2)** 819 **3)** 720 **4)** 601

63. The locus of the vertices of the family of parabolas $y=\frac{a^3x^2}{3}+\frac{a^2x}{2}-2a$ is $(a\in R)$

1) $xy=\frac{105}{64}$ **2)** $xy=\frac{3}{4}$ **3)** $xy=\frac{35}{16}$ **4)** $xy=\frac{64}{105}$

64. The value of $\left(4\cos^2 9^0-1\right)\left(4\cos^2 27^0-1\right)\left(4\cos^2 81^0-1\right)\left(4\cos^2 243^0-1\right)$ is

1) -1 **2)** 1 **3)** 2 **4)** -2

65. Let A, B, C be 3×3 matrices such that B, C are Skew-symmetric and A is symmetric. Consider the statements.

(S1) $A^{43}B^{64}-B^{64}A^{43}$ is Symmetric

(S2) $A^{64}B^{43}-B^{43}A^{64}$ is Symmetric Then,

1) Statement 1 is false and statement 2 is true

2) Statement 1 and 2 both are false

3) Statement 1 and 2 both are true

4) Statement 1 is true and statement 2 is false

66. Let the function $f:[0,2]\to\mathbb{R}$ be defined by $f(x)=\frac{4^x}{4^x+4}$. Then the value of $f\left(\frac{1}{2024}\right)+f\left(\frac{2}{2024}\right)+f\left(\frac{3}{2024}\right)+.....f\left(\frac{4046}{2024}\right)+f\left(\frac{4047}{2024}\right)$ is _____

1) 2023 **2)** $\frac{4047}{2}$ **3)** 2024 **4)** $\frac{4049}{2}$

67. $\lim\limits_{x\to 2}\left[\left(\frac{x^3-4x}{x^3-8}\right)^{-1}-\left(\frac{x+\sqrt{2x}}{x-2}-\frac{\sqrt{2}}{\sqrt{x}-\sqrt{2}}\right)^{-1}\right]$ is equal to

1) $\frac{1}{2}$ **2)** 2 **3)** 1 **4)** $\frac{2}{3}$

68. If Z is complex number such that $\frac{z-i}{z-1}$ is purely imaginary then maximum value of $|z-(3+3i)|$ is

1) $\sqrt{2}$ **2)** $3\sqrt{2}$ **3)** $2\sqrt{2}$ **4)** $4\sqrt{2}$

69. If $P=\begin{bmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{bmatrix}, A=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ and $Q=PAP^T$, then $P^T\left(Q^{2005}\right)P$ is equal to

1) $\begin{bmatrix}1 & 2005\\ 0 & 1\end{bmatrix}$ **2)** $\begin{bmatrix}\frac{\sqrt{3}}{2} & 2005\\ 1 & 0\end{bmatrix}$ **3)** $\begin{bmatrix}1 & 2005\\ \frac{\sqrt{3}}{2} & 1\end{bmatrix}$ **4)** $\begin{bmatrix}1 & \frac{\sqrt{3}}{2}\\ 0 & 2005\end{bmatrix}$

70. Let $x^2+y^2+Ax+By+C=0$ be a circle passing through $(0,6)$ and touching the parabola $y=x^2$ at $(2,4)$. Then, $A+B+C$ is equal to _____.

1) 16 **2)** $\frac{-36}{5}$ **3)** $\frac{36}{5}$ **4)** -8

71. The function $f(x)=\begin{cases}\frac{\pi}{4}+\tan^{-1}x, & |x|\le 1\\ \frac{1}{2}(|x|-1), & |x|>1\end{cases}$ is

1) both continuous and differentiable on $R-\{-1\}$

2) both continuous and differentiable on $R-\{1\}$

3) continuous on $R-\{1\}$ and differentiable on $R-\{-1,1\}$

4) continuous on $R-\{-1\}$ and differentiable on $R-\{-1,1\}$

72. For the function $f(x)=4\log_e(x-1)-2x^2+4x+5,\ x>1$ Which one of the following option is not correct?

1) f is increasing in (1,2) and decreasing in $(2,\infty)$

2) $f(x)=-1$ has exactly two solutions

3) $f'(4)-f''(2)>0$

4) $f(x)=0$ has a real root in the interval $(1,2)$

73. The value of the integral $\int_{1/2}^{2} \frac{\tan^{-1}x}{x}dx$ is equal to

1) $\frac{1}{2}\log_e 2$ **2)** $\pi\log_e 2$ **3)** $\frac{\pi}{4}\log_e 2$ **4)** $\frac{\pi}{2}\log_e 2$

74. If S be the set of all values of λ, for which the shortest distance between the lines

$\frac{x-\lambda}{0}=\frac{y-3}{4}=\frac{z+6}{1}$ and $\frac{x+\lambda}{3}=\frac{y}{-4}=\frac{z-6}{0}$ is 13 then the value of $4\left|\sum_{\lambda\in S}\lambda\right|$ is equal to

1) 306 **2)** 153 **3)** 308 **4)** 304

75. The system of Linear equations

$\lambda x+2y+2z=5$,

$2\lambda x+3y+5z=8$,

$4x+\lambda y+6z=10$ has

1) No Solutions when $\lambda=8$

2) a Unique Solution when $\lambda=-8$

3) infinitely many Solutions when $\lambda=2$

4) no Solutions when $\lambda=2$

76. Let α *and* β be the roots of $x^2-8x-2=0$. If $a_n=\alpha^n-\beta^n$ for $n\geq 1$, then the value of $\frac{a_{10}-2a_8}{4a_9}$ is

1) 2 **2)**1 **3)** 4 **4)** 3

77. Let $f:R\to R$ be a function defined by $f(x)=x^3+x-1$ and 'g' is the inverse of f, then the value of $\frac{g(9)}{g'(9)}$ is (g' is the derivative of g)

1) 26 **2)** 32 **3)** 11 **4)** 17

78. A signal which can be green or red with probability $\frac{4}{5}$ and $\frac{1}{5}$ respectively, is received by station A and then transmitted to the station B. The probability of each station receiving the signal correctly is $\frac{3}{4}$. If the signal received at station B is green, then the probability that the original signal was green is

1) $\frac{20}{23}$ **2)** $\frac{9}{20}$ **3)** $\frac{4}{13}$ **4)** $\frac{6}{7}$

79. Let $A=\{x\in R:[x+3]+[x+4]\le 1\}, B=\left\{x\in R:3^x\left(\sum_{r=1}^{\infty}\frac{3}{10^r}\right)^{x-3}<3^{-3x}\right\}$, where [t] denotes greatest integer function. Then,

1) $A\subset B$ **2)** $A\cap B=\phi$ **3)** $A=B$ **4)** $B\subset A$

80. If $S_n=\sum_{r=1}^{n} r!$ then for $n>6\left(given\sum_{r=1}^{6} r!=873\right)$

	COLUMN-I		COLUMN-II
A)	$\sin^{-1}\left(\sin\left(S_n-7\left[\frac{S_n}{7}\right]\right)\right)$	**P)**	$5-2\pi$
B)	$\cos^{-1}\left(\cos\left(S_n-7\left[\frac{S_n}{7}\right]\right)\right)$	**Q)**	$2\pi-5$
C)	$\tan^{-1}\left(\tan\left(S_n-7\left[\frac{S_n}{7}\right]\right)\right)$	**R)**	$6-2\pi$
D)	$\cot^{-1}\left(\cot\left(S_n-7\left[\frac{S_n}{7}\right]\right)\right)$	**S)**	$5-\pi$
		T)	$\pi-4$

(where [ ] denotes greatest integer function)
The Correct match:

1) $A-P;B-Q;C-P;D-S$ **2)** $A-Q;B-R;C-P;D-T$

3) $A-R;B-S;C-T;D-S$ **4)** $A-P;B-S;C-Q;D-T$

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Marking scheme: +4 for correct answer, 0 if **not attempt** and **-1** in all other cases.

81. If $\sin^{-1}\left(x-\frac{x^2}{2}+\frac{x^3}{4}-.....\right)+\cos^{-1}\left(x^2-\frac{x^4}{2}+\frac{x^6}{4}-....\right)=\frac{\pi}{2}$ for $|x|<\sqrt{2}$ then number of values of x equal to

82. Let k be a positive real number and $A=\begin{bmatrix}2k-1 & 2\sqrt{k} & 2\sqrt{k}\\ 2\sqrt{k} & 1 & -2k\\ -2\sqrt{k} & 2k & -1\end{bmatrix}, B=\begin{bmatrix}0 & 2k-1 & \sqrt{k}\\ 1-2k & 0 & 2\sqrt{k}\\ -\sqrt{k} & -2\sqrt{k} & 0\end{bmatrix}$

If $\det\left(adj\left(adjA\right)\right)+\det\left(adjB\right)=10^{12}$, then $\left[k\right]$ is equal to _____(where [.] is greatest integer function)

83. Let $y = y(x)$, be the solution of the differential equation $(x-1)\frac{dy}{dx} + 2xy = \frac{1}{x-1}$, with $y(0) = 1$. If $y(2) = \frac{e^{\lambda}+1}{\alpha e^{\lambda}}$ then $\lambda - \alpha =$

84. Coefficient of x^4 in $\left(1 + x + x^2 + x^3\right)^9$ is

85. The total number of ordered pairs (x, y) which satisfies $y = |\sin x|, y = \cos^{-1}(\cos x)$ where $-2\pi \le x \le \pi$

86. Set A has m elements and Set B has n elements. If the total number of subsets of A is 96 more than the total number of subsets of B, then the value of $\left[\frac{m+n}{m-n}\right]$ = _____. (where [.] is greatest integer function)

87. Let $X_1, X_2, ...X_{18}$ be eighteen observations such that $\sum_{i=1}^{18}(X_i - \alpha) = 36$ and $\sum_{i=1}^{18}(X_i - \beta)^2 = 90$, where α and β are distinct real numbers. If the standard deviation of these observations is 1, then the value of $|\alpha - \beta|$ is _____.

88. If the line $l_1 : 3y - 2x = 3$ is the angular bisector of the lines $l_2 : x - y + 1 = 0$ and $l_3 : \alpha x + \beta y + 17 = 0$, then $\alpha^2 + \beta^2$ is equal to _____.

89. Let $f : [-3,1] \to R$ be given as

$$f(x) = \begin{cases} \min\{(x+6), x^2\}, & -3 \le x \le 0 \\ max\{\sqrt{x}, x^2\}, & 0 < x \le 1 \end{cases}$$

If the area bounded by $y = f(x)$ and x-axis from $x = -3$ to $x = 1$ is A, then the value of $[A]$ is equal to__(where [.] is greatest integer function)

90. Let three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ be such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}, \vec{a} \cdot \vec{c} = 7$ and $\vec{b}$ is perpendicular to $\vec{c}$, where $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{k}$, then the value of $2|\vec{b} + \vec{c}|^2$ is_____.